DIMENSIONAL STONE MINING

1.1 DIMENSIONAL STONE

Dimension stone is defined as natural rock material quarried for the purpose of obtaining blocks or slabs that meet specifications as to size (width, length, and thickness) and shape suitable for use as building stone, ornamental stone, and monumental stone.

Majority of rocks commonly used as dimensional stone are grouped into six major commercial categories

i. Granite	**ii. Black Granite**	**iii Sandstone**
iv. Limestone	**v. Marble and**	**vi.Slate.**

These are again subdivided into other stones such as grey granite, quartzite and quartzite sandstone, and green marble.

Other categories used for commercial purposes include **bluestone**, **multicolor basalt or traprock, flagstone and greenstone**. In addition there are number of less commonly used rock types in the dimensional stone industry that fall outside the normal commercial categories.

Dimension stone is mined by more **precise and delicate techniques**, such as diamond wire saws, diamond belt saws, Jet-piercers, or light and selective blasting with Primacord, a weak explosive.

Dimension stone is used as **countertops and bathroom vanities** both involve a finished slab of stone. The Standard thicknesses of dimensional stone are 2 cm or 3 cm. Often 2 cm slabs are laminated at the edge to create the appearance of a thicker edge profile. The slabs are cut to fit the top of the kitchen or bathroom cabinet, by measuring, templating or digital templating.

Countertop slabs are commonly sawn from rough blocks of stone by reciprocating gangsaws using steel shot as abrasive. More modern technology utilizes diamond wire saws which use less water and energy. Multi-wire saws with as many as 60 wires can slab a block in less than two hours. The slabs are finished i.e. polished, honed, then sealed with resin to fill micro-fissures and surface imperfections typically due to the loss of poorly bonded elements such as biotite.

The fabricators shop cuts these slabs down to final size and finishes the edges with equipment such as hand-held routers, grinders, CNC equipment, or polishers. In 2008, concerns were raised regarding radon emissions from granite countertops in US. The National Safety Council states that the contributions of radon to inside air come from the soil and rock around the residence (69%), the outdoor air and the water supply (28%), and only 2.5% from all building materials-including granite countertops.

The stone for countertops or vanities is usually granite, but often is marble (especially for vanity tops), and is sometimes limestone or slate. The majority of the stone for this application is produced in **Brazil, Italy, and China**.

There are a number of smaller applications for **buildings and traffic-related uses**. Building components include stone used as **exterior veneer, ashlar, or other shapes**. Veneer is a non-load bearing facing of stone attached to a backing, of an ornamental nature though it protects and insulates. Ashlar is a square block of stone, often brick-sized, for facing of walls primarily exterior.

The other shapes are rectangular blocks used for stair treads, sills, and coping (coping is sometimes non-rectangular). The shapes subject to foot traffic will usually have an abrasive finish such as honed or sandblasted. The stone is mostly limestone, but often is quartz-based stone such as sandstone, or even marble or granite. Roofing slate is a thin-split shingle-sized piece of slate, and when in place forms the most permanent kind of roof; slate is also used as countertops and flooring tile.

Traffic-related stone is that which is used for **vehicular curbing and pedestrian flagstone**. Curbing is thin stone slabs used along streets or highways to maintain the integrity of sidewalks and borders. Flagstone is a shallow naturally irregular-edged slab of stone, sometimes sawed into a rectangular shape, used as paving (almost always pedestrian). For curbing, the stone is almost always granite, and for flagstone the stone is almost always quartz-based stone such as sandstone and quartzite.

2.0 Mining of Dimensional Stones

Mining of dimensional stone is quite different from conventional mining. Here the blocks of different sizes in cuboidal form are extracted with due care avoiding cracks and damages to the mined out blocks using specialised techniques.

Dimensional Stone mining in India can be classified into three categories namely, manual, semi-mechanised and fully mechanised method.

Steps involved in extraction of dimensional stone are

a) removal of overburden to expose the minable block
b) extraction of primary block
c) size reduction of primary block to secondary block.
d) handling and loading of blocks and
e) transportation.

2.1 REMOVAL OF OVERBURDEN TO EXPOSE MINABLE BLOCK

The removal of overburden is necessary to expose the mineral present beneath it. Top soil is removed with the help of backhoe, remaining hard strata is blasted off with the use of explosives. Compressed air drills are widely used to drill holes in the overburden. The waste rock is transported to a distance sufficient far away from the quarry where no reserve of dimensional stone exist underneath. The overburden is handled normally using a combination of hydraulic excavator and dumper. Some of the mines utilize a combination of backhoe and tipper. Widely used class of dumpers are 35 tonner Euclid EH600 dumpers of Tata Hitachi make. Hydraulic backhoe type excavator of 2.4 m^3 bucket capacity are most common in marble mines of Rajasthan.

Figure 1: Photograph showing overburden dumping site at R K Marbles Pvt Ltd (Rajasthan)

Overburden which contains a large amount of stone that can be utilised by local sculptures and small scale tile makers are sold to customers at reasonable price. Waste material containing small pieces of stone, marble, clay and soil are then dumped at dump yards (Figure 1).

2.2 EXTRACTION OF PRIMARY BLOCK

Cuboidal shaped rock normally of size 10 m × 10 m × 3 m or 15 m × 6 m × 6-9 m are extracted as primary blocks. These blocks are separated from the *in-situ* rock using diamond wire saw. In order to cut a primary block on a bench either four planes or three planes are necessary depending upon situation. If no primary block is extracted from a bench then for cutting the first block a minimum of four planes are required to be cut, three in vertical plane out of which two are in lateral direction and one on the back side of the block, the fourth plane is a horizontal plane at the bottom of the block (Figure 2). This initial cut is known as gully or blind cut. Subsequent extraction of blocks then would require only three planes to be cut, two of which are vertical planes one of them being in lateral direction and other on the back side of the

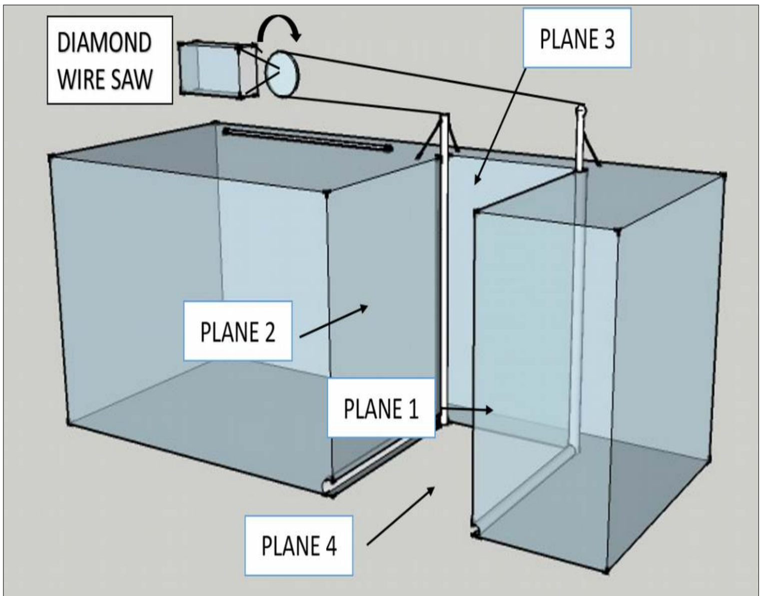

Figure 2: Figure showing minimum of four planes to be cut in a situation when no primary block has been extracted from a face.

block. The third plane will be in horizontal direction at the bottom of the block (Figure 3).

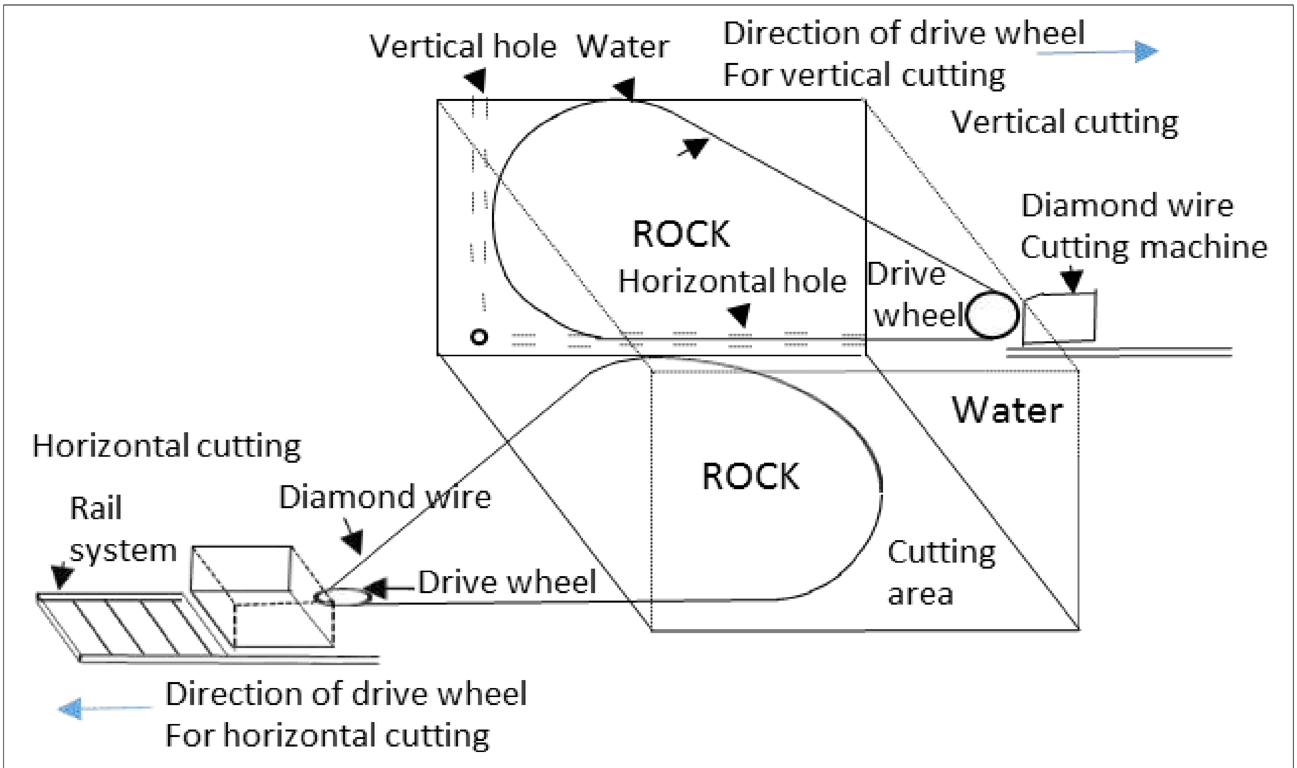

Figure 3: Schematic diagram of vertical and horizontal cutting with the help of Diamond Wire

2.2.1 GULLY CUT

Initially when no block has been extracted from a freshly developed bench, the first cut on this bench would require four cuts as mentioned above. Cutting of face on the backside of the block is carried out with the help of diamond wire saw having special arrangement made of channels and a set of pulley carrying diamond wire (Figure 4 a&b).

Figure 4a: Photograph of gully cut at D J Nellum Pvt Ltd, Agaria (Rajasthan).

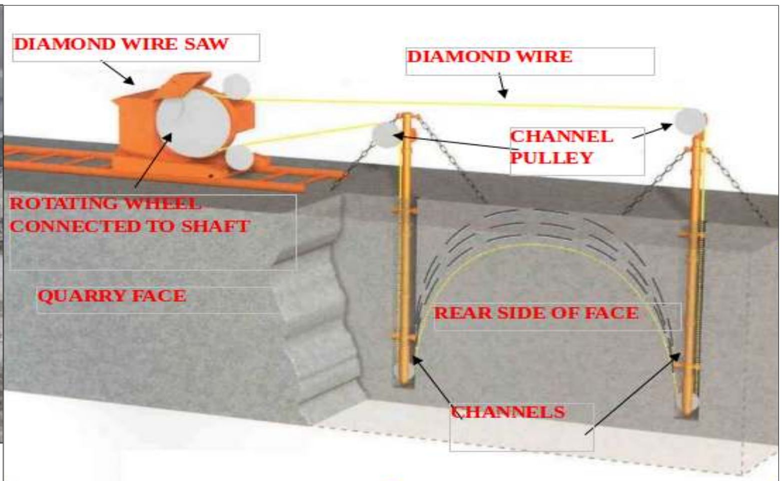

Figure 4b: Schematic diagram of Blind cut technique (courtesy R K Marbles Pvt Ltd, Morwad)

This is an innovative technique for development of quarry front, normally referred to as blind cut or gully cut. Initially when this technique was not available, drilling and blasting method was used, which not only wasted the mineable block but also induced undesirable cracks in adjacent rock body. This technique utilises two vertical holes of diameter 190 mm drilled up to a desired depth.

These holes are then connected to horizontal holes which serve the purpose of drainage of slurry as the cutting progresses. The channel columns with pulleys at both ends carrying a loop of diamond wire are then lowered in the vertical holes. The loop of the diamond wire is then fed over the flywheel of the diamond wires sawing machine. Gully cut is made in five to six stages of blind cut, taking out the block from front of the face.

2.2.2 DRILLING OPERATION

Drilling is one of the most important steps in extraction of dimensional stones. Various drill machines such as Slim Drill, Quarry Master, Quarry Voyager, LM-100 of Atlas Copco make are popular in dimensional stone mining. The purpose of these machines is to drill coplanar holes used for passage of diamond wire. With improvements in bit design and sturdy frame over a period of time these drill machines have outperformed the pace of drilling. Description of the drilling rate of respective drilling equipment is given in Table 1.

2.2.2.1 VERTICAL DRILLING

The vertical drilling is part of making the coplanar holes to pass the diamond cutting wire through it. This operation is carried out by LM-100 (Atlas Copco) drill machine. It is a pneumatic powered drilled machine. The diameter of the drill bit used is 110 mm, in which 14 button bits are mounted and 2 opening are available for outlet of the air. This machine is operated by compressed air. Compressors of make Chicago Pneumatics are very popular in these area. Smaller quarries use slim drills erected on a base frame. These slim drills are pneumatically operated machines.

Table 1: Drilling performance at R. K. Marbles Pvt. Ltd

Drill machine	Hole diameter (mm)	Depth of hole (m)	Time taken to drill(minutes)	Drilling rate (m/hr.)	Rock /mineral
LM-100 (Atlas Copco)	110	3	19.50	9.23	Sandstone
		1.2	7.10	10.14	
DC-120 (Sandvik)	34	1.6	2.82	34.04	Marble
			2.58	37.20	
			2.50	38.4	
Quarry voyager (Marini equipments)	110	9	70	7.71	Marble

2.2.2.2 HORIZONTAL DRILLING

Quarry voyager of make Marini equipment is used for horizontal drilling. Slim drill were used in past in most of the mines. However, over time due to various advantages like faster drilling rate and sturdy design, Quarry Voyager (Figure 5) have replaced the slim drill . Horizontal spacing between the holes is about 5 to 6 feet. Drilling rate of the machine is 12 cm/min. Vertical drilling is part of making the coplanar holes to pass the diamond wire through it. This operation can also be is carried out by LM-100 (Atlas Copco) drill machine. It is a pneumatic drill machine. The diameter of the drill bit is 110 mm, in which 14 button bits are mounted and two openings are available for air outlet. This machine is generally used in conjunction with pneumatic compressor of make Chicago Pneumatics.

Figure 5: Photograph showing quarry voyager drilling a horizontal hole at face (D J Nellum Pvt Ltd, Agaria)

2.2.3 MATCHING OF HOLES

Based on Production planning, holes are marked on rock surface. There are two methods of matching holes. The first one being manual matching of holes and the second being the use of Cerca Fori. Both these method are described below in the following subsections.

2.2.3.1 MANUAL METHOD OF MATCHING HOLE

Horizontal holes are marked with respect to vertical holes by shovel, plumb bob and manual eye estimation. However latest innovation such as the use of dot laser machine are also in vogue in most of the mines.

2.2.3.2 MATCHING OF HOLES USING CERCA FORI

Cerca fori is an instrument used to change the alignment of coplanar holes with more than 90% accuracy. This instrument consist of a transmitter, receiver and a display unit. The instrument works on the principle of propagation of electromagnetic waves that can pass through rock medium. The instrument consist of two rods, at the end of one of the rod an electromagnetic wave transmitter is fitted and at one end of other

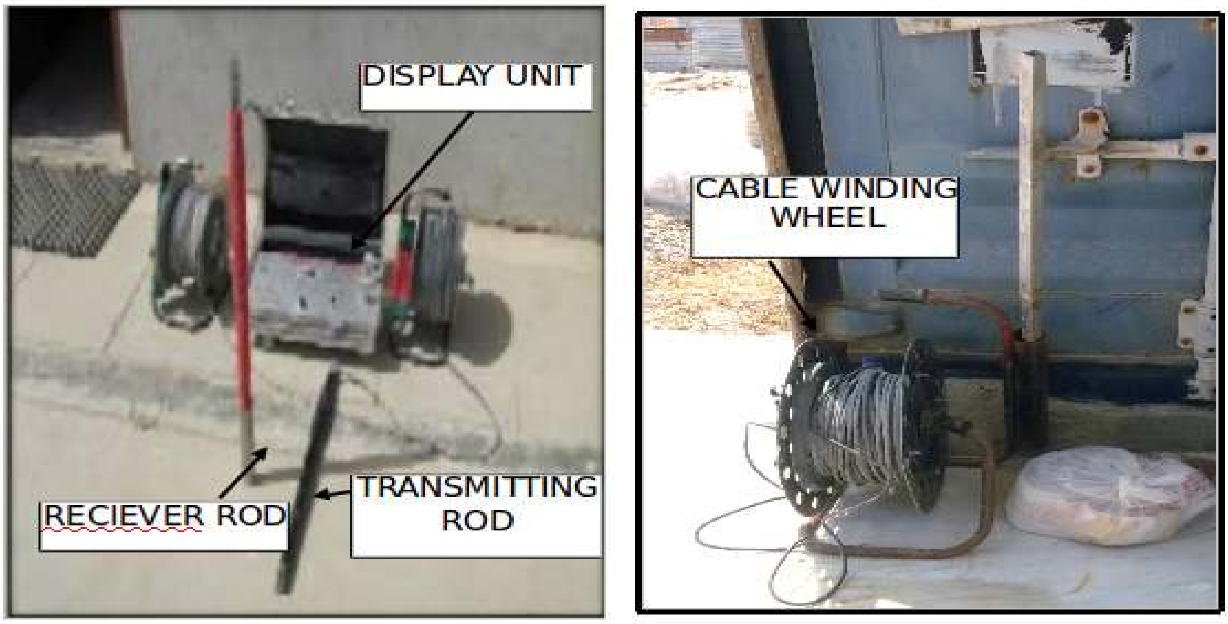

Figure 6: Photograph showing cerca fori and its cable winding wheel

rod a receiver is fixed. The other ends of the rods are fitted with cables connected with the display unit (Figure 6). Now in order to find out the alignment of the mutually perpendicular holes, the transmitter probe is inserted in one hole and the receiver probe is inserted in other hole. The transmitter and receiver are then triggered. If the holes are not matched, the indicator on the display unit will display the direction of mismatch i.e. mismatch in either right or left side of one of the holes. Accordingly the driller has to make a decision to re-drill in the direction to match the hole. The system is cost effective and saves time in comparison with conventional method.

2.2.4 CUTTING BY DIAMOND WIRE SAW

Diamond wire saw is one of the most critical machinery used in marble cutting. Its advantage lies in its extreme implicitly and incredible performance. This machine cuts the marble directly from the quarry face with a long steel cable fitted with a diamond cutting element (Figure 7 & 8) as cutting media which a kept cool by continuous flow of water. The water also clear away the cutting debris. Impregnated beads of size 10-

11mm in diameter are used. One meter length of cable consist of 33-40 beads. The bead density varies according to the requirement of cutting rate and the deposit characteristics. The suitability of the bead density is at present selected on trial and error estimates.

In order to cut a block of marble, the first requirement is to drill co-planer holes and then insert diamond wire in the holes. In order to achieve this, a small diameter steel wire is tied to a jute cloth and the jute cloth is inserted in the hole tightly then water is pressurized in the hole. Due to the pressure of water the jute cloth travel through the hole and come out from other end. Then the diamond wire is tied to this small diameter wire already in hole and it is pulled from other end. The two end are kept collinear to each other and a steel coupling is crimped at these ends to make it an endless loop.

Diamond wire saw machine is mounted on parallel steel pipes separated by a distance of 60 cm. One of the rail pipe contains rake over which the pinion of the machine rotates (Figure 9). This helps the machine to travel back as the cutting through the wire saw progresses. A proper tension is provided to the diamond wire and main motor is turned on. In this way faces are cut from all sides.

There are three variant of this machine based on power requirement. They are 60 HP, 40 HP and 20 HP machines. The higher rated machines i.e. 60 HP and 40 HP are used for primary block cutting whereas other i.e. 20 HP is used for horizontal and secondary cutting. Diamond wire saw used at R. K. marble mine was manufactured by Raj Process and Optima Machineries. The average cutting rate of diamond wire saw is 55.12 sq.ft/hr. These machines are semi portable which can be transported easily inside the quarry on trucks (Figure 10).

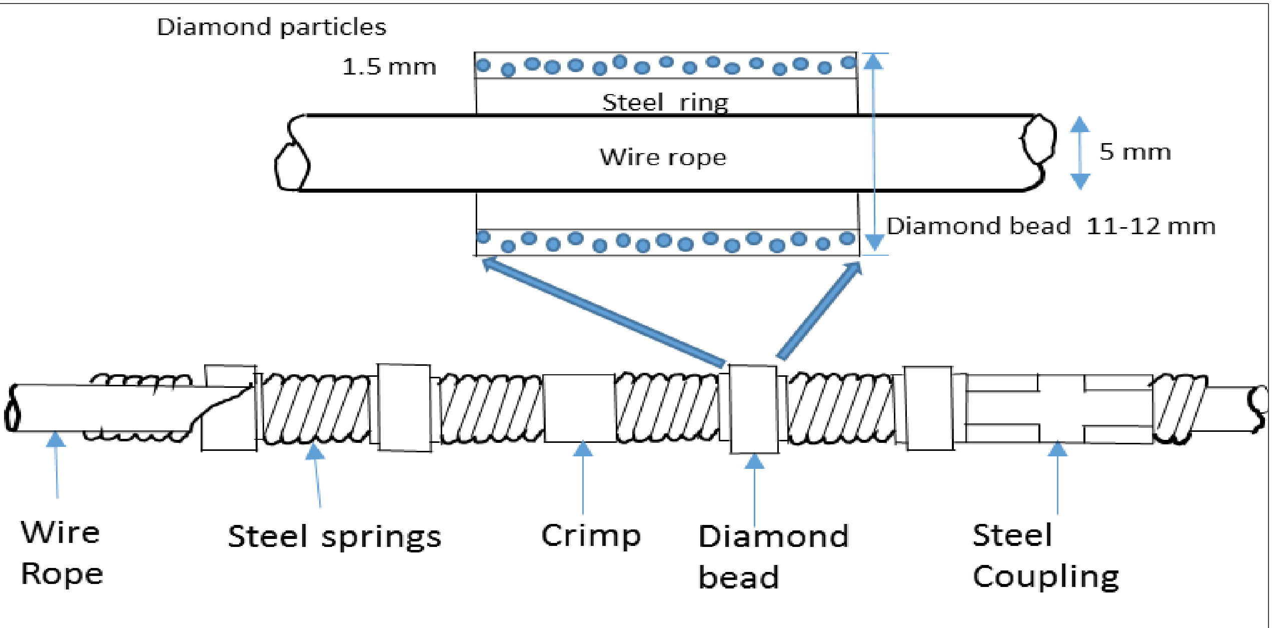

Figure 7: Schematic diagram of various elements used in diamond wire

Figure 8: Photograph showing diamond wire saw at Agaria marbles (D J Neelum Pvt Ltd, Agaria).

Figure 9: Photograph showing rake and pinion arrangement in tracks of diamond wire saw.

Figure 10: Truck used for transportation of wire saw machine

2.2.5 USE OF HYDRAULIC AND PNEUMATIC BAG FOR SEPARATION OF BLOCK

Application of hydraulic and pneumatic bags are simple efficient way to create a small gap between the cut created by diamond wire saw. This help to insert backhoe teeth or

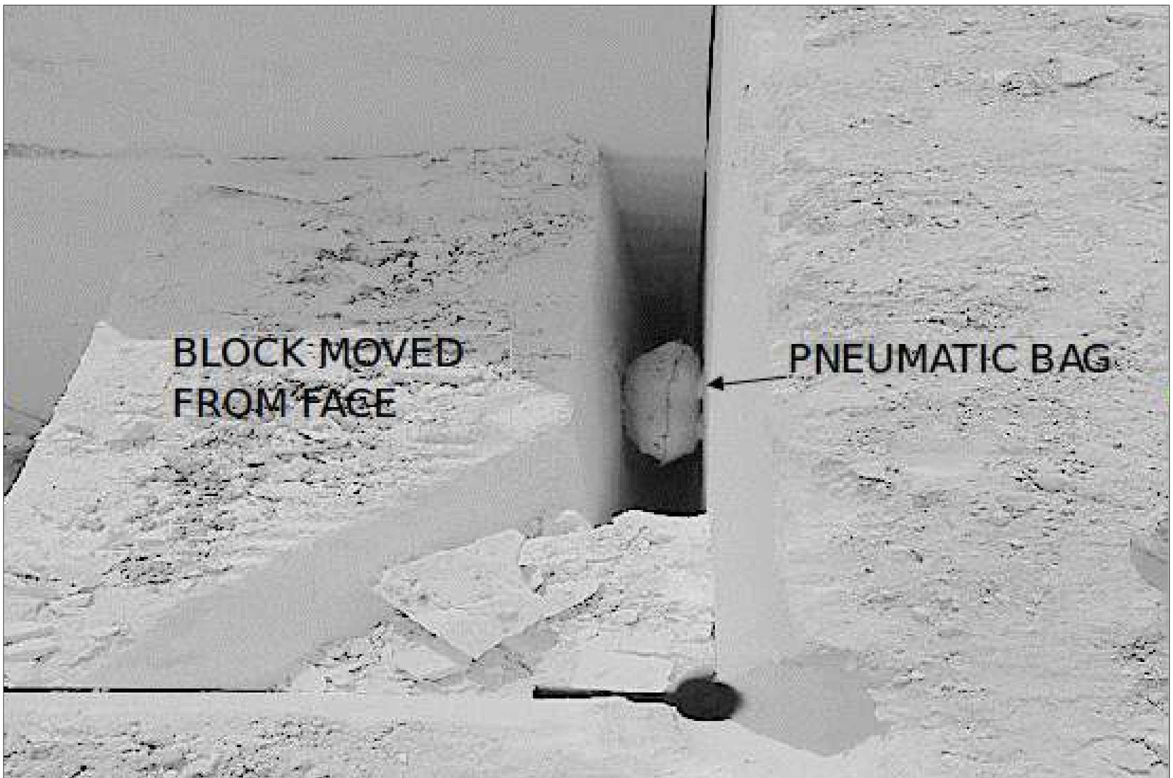

Figure 11: Photograph showing gap created by pneumatic bag

pusher arm edge to move in and push the block to topple over the debris of waste rock. These bags are made from “cold rolled cold annealed” high tensile steel sheet. Two pieces of 1 m ×1 m are welded at margin leaving an edge of 10-12 mm. Water with pressure of 3 MPa is filled with the help of positive displacement pump, which exerts pressure on the walls. In the same way, pneumatic bags having slightly larger dimension 1 m×1.5 m are inflated by pumping compressed air (Figure 11).

2.2.6 OVERTURNING AND TOPPLING OF BLOCK

Once the mega block is made free from in-situ rock with the help of wire saw cut at the bottom, side and rear, toppling process of mega block comes in to picture. It is achieved either by the use of the bucket of an excavator (Figure 12) or a pusher arm attached on a wheel loader (Figure 13). Pusher arm is a recent development in block toppling.

Figure 12: Photograph showing backhoe bucket nose for toppling of block at R K Marbles Pvt Ltd.

Figure 13: Photograph showing pusher arm for toppling of block at R K Marbles Pvt Ltd.

2.3 SECONDARY DRILLING AND SPLITTING

Secondary drilling is required to divide the primary blocks into smaller blocks of desired dimensions. Sandvik DC-120 is used for line drilling (Figure 14). It is a tyre mounted, diesel propelled vehicle. This drilled machine is also fitted with 2 HP motor for dust suction system. It collect all the dust to its reservoir and can be emptied when required. Diameter of the bit is 35 mm. The main motor runs at 1500 rpm with a 37 kW power output .This generates enough power to drill at the rate of 120 ft. /hr. The drill rod is tapered at the end at an angle of 12 degrees where the bit can be attached. This tapering of the rod is provided to ensure proper fitment of drill bit. After secondary drilling primary block is released from a small height to fall freely on ground, this imparts sufficient facture plane along the drill lines. This way the secondary blocks are seperated from the primary block.

Wedge and feathers are also used to separate the secondary blocks. It generally consist of wedge (plug) with two feathers (shims). The plug is placed in between the shims such that when the plug is hammered it create a crack joining all the holes. (Figure 15).

Figure 14: Photograph of Line drilling machine DC-120 (SANDVIK)

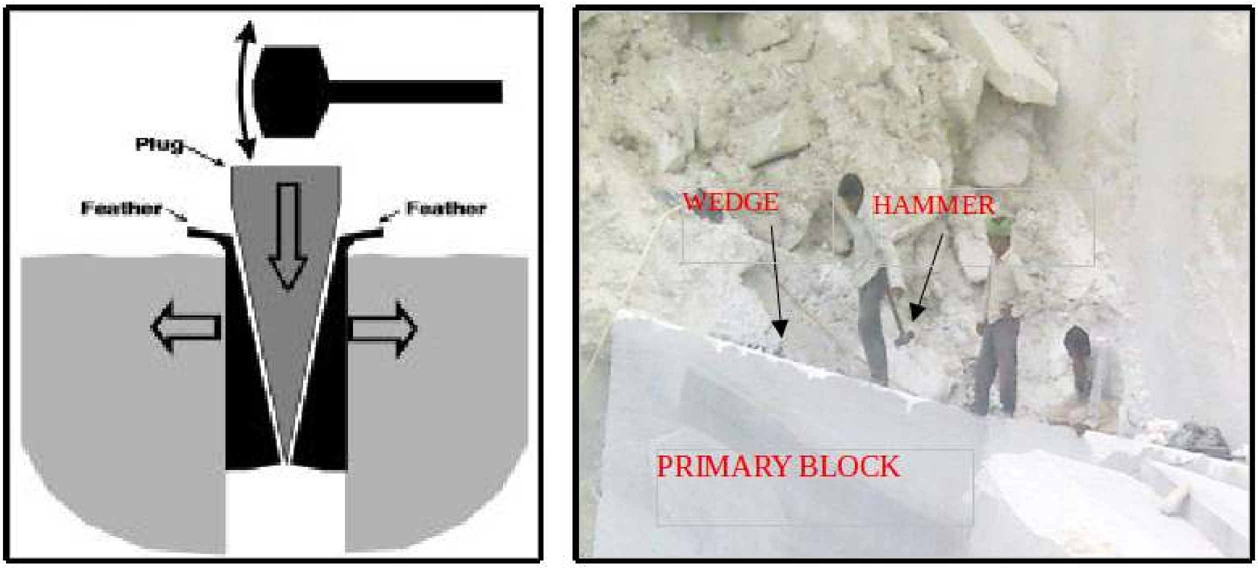

Figure 15: Schematic diagram of Wedge and feather

2.4 LOADING AND TRANSPORTATION

In conventional mining methods the material is transported with the help of dumpers which carries it to the pit top. But in case of marble quarries, the secondary block is lifted with the help of a specially designed front end loaders having rock pallet fork lift arrangement. Another method is to lift the block is with the help derrick crane. These are described in following subsections.

2.4.1 DERRICK CRANE

Derrick cranes are used for loading blocks and shifting if from pit bottom to pit top. It essentially consists of a vertical mast which is stabilized with the help of guy lines. These guy lines can be either cables or stiff legs. The base of the mast consists of an arrangement for swinging the boom as well as the mast in lateral direction (Figure 16). The swing angle of the boom is restricted to less than 180° because of the presence of the guys or stiff legs. The boom lift can be controlled through boom lines or topping lines. Boom lift and lowering are not carried out frequently. It is done only when the face changes. To lift a load a separate line runs up and over the mast with the hook on its end much similar to an ordinary crane.

A view of cranes in use at Odwas and Agaria mines are shown in Figure 17 & 18. Derrick cranes are specified by to its safe working load and boom length. For example a 50t/80 m describes a derrick crane which can handle a block weighing up to 50 tonnes and the boom length of the crane is 80 m.

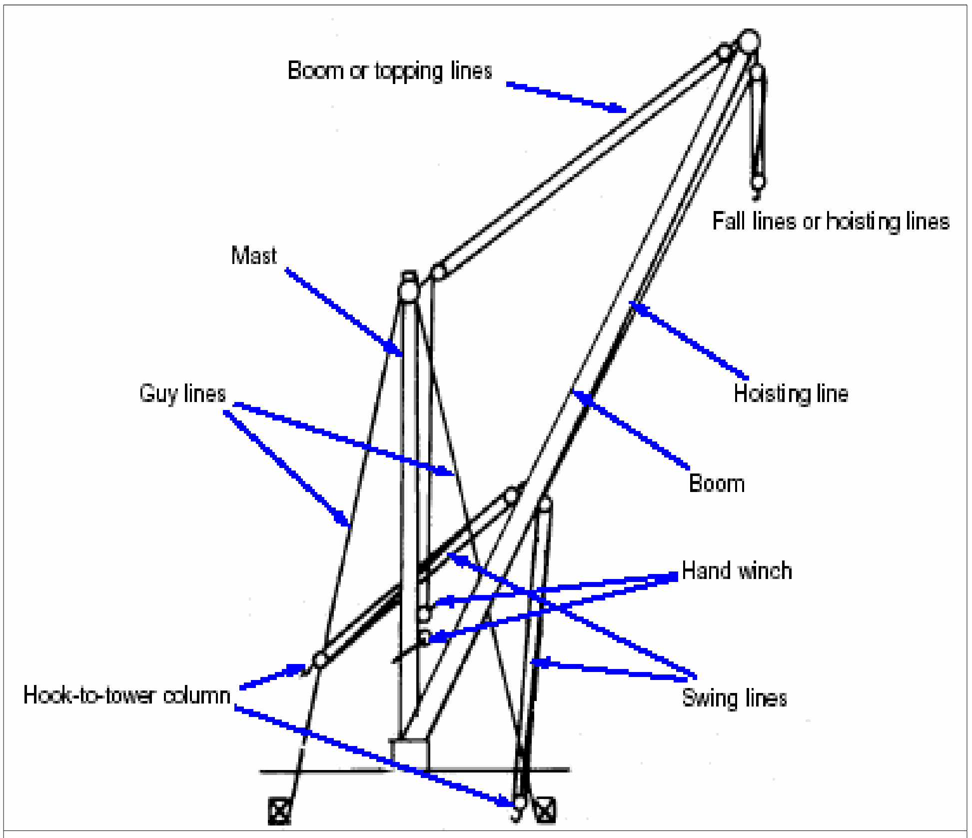

Figure 16: Schematic diagram of a Derrick crane

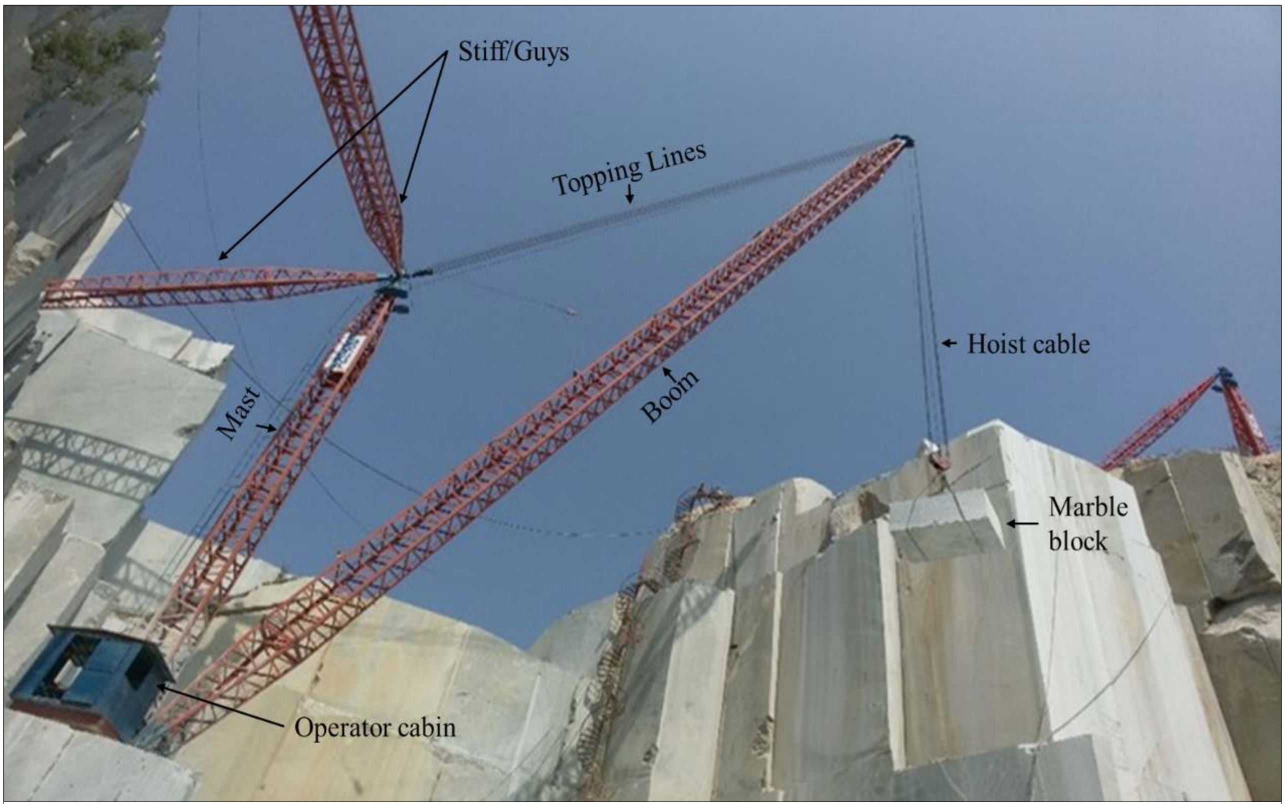

Figure 17: Photograph showing derrick crane at D J Neleum Pvt Ltd, Odwas.

Figure 18: Base of a derrick crane

2.4.2 BLOCK HANDLER

A block handler is diesel propelled vehicle consisting of an arrangement known as rock pallet fork arrangement used to lift the marble blocks. Generally the tipping load varies from 12 to 22 tonnes (Figure 19).

Figure 19: Photograph showing CAT-980G Block Handler

2.5 PROCESSING

After mining of the blocks from quarry, these blocks are transported to processing plant make slabs of suitable dimension. The steps followed in processing of marble blocks are as follows:

a) Cutting of locks into slabs by the use of gang saw.

b) Meshing to provide support to the slab for subsequent processing.

c) Heating to reinforce the glue between mesh and slab.

d) Rough grinding of marble slabs.

e) Quick heating for effective propagation of epoxy resin into the hair line cracks

Gang saw is a widely used machine to cut block into thick slabs (Figure 20). It consist of diamond impregnated segment that are brazed at equally spaced interval known as pitch. Gang saws are classified into two types according to the movement of the block or the movement of the saw frame. They are a) Frame lowering machine b) Trolley rising machine.

In frame lowering machine the frame containing saw blades moves down to provide thrust on the block while cutting. Alternatively the frame of the saw is stationary and the trolley, over which the block is placed, is moved upwards to impart the thrust on the block. Such machines are known as trolley rising machines. Frame lowering machines are widely used in Rajasthan. The specification of multi-bladed gang saw is given in Table 2.

Table 2: Specifications of multi-blade saw/gang saw

Length of blade	4100/4200*180mm
Thickness of blades	2.5 mm
Rotation speed of driving shaft	105 rpm
Stroke length	600 mm
Tensioning of blades	10-15 tonnes per blade
Tensioning device of blade	Mechanical
Water consumption	10 liter / blade
Lubricating system	Self
Main blade motion	To & fro sliding motion
Cut down feed speed	4-5 inch/hr.
Main motor power	110kW
Cut down feed motor power	3 HP
Fast lowering motor power	7.5 Kw
Total installed power	132 kW

Figure 20: Photograph showing gang saw operation at D J Marble Processing Plant

After cutting of the block the next requirement is to polish the slabs. However, before polishing the slab, certain reinforcement is necessary to prevent it from breaking. This is done through meshing, a process in which glass fibre mesh is glued with the help of epoxy resin at the back of the slab. Some cracks may also be available at the corners of the slab. In order to prevent the side cracks to propagate inside the slab, small strips of tiles are glued around the edges with the help of epoxy resin for reinforcement of the slab.

The entire meshed slab is heated inside a chamber or by use of hot air blowers. This heating chamber can be either of tower type in which multiple slabs are heated at a time (Figure 22). The other arrangement can be of single slab heating process (Figure 23). The process of heating actuates the polymerisation of epoxy.

After this robots are used to handle the slabs which consist of suction cups (Figure 24), to keep slabs on automatic line polish machine (Pedrini, Italy). This machine is having 16 different heads with brush. A pressure from 5 to 15 bar is adjusted at different heads. The average polishing time is around half hour/slab. It is equipped with sensor to detect the slab smoothness. Water is used as lubricant to resist it from excessive wear of brush heads and better polishing of tiles. The different polish ranges from horned to antique polish. After polishing, a layer of epoxy is spread over the top to fill the remaining hairline cracks. Again it is quick heated to 65 ℃ . After this the tiles are ready to be sold in market.

Figure 21: Photograph showing gantry crane is used for handling of blocks at processing plant.

Figure 22: Photograph showing automatic slab heating tower at processing plant (D J Marble Pvt Ltd)


Figure 23: Photograph showing single slab heating machine at processing plant (D J Marble Pvt Ltd)

Plate 2.20: Photograph showing 7×4 cup suction type automatic slab handling robot.